

Statistics Practice Set — S22

S22 will use **vertical datasets only**.

The difficulty is slightly higher than S21: focus on **multi-step calculations + interpretation + choosing the correct statistical measure**.

Q1 — Mean vs Median + Outlier Investigation

Order Value

600

650

700

750

800

850

900

5500

Answer:

1. Calculate the mean.
 2. Calculate the median.
 3. Which measure better represents typical order value?
 4. Is the distribution likely right-skewed or left-skewed?
 5. Why is the mean much higher than the median?
 6. Does the mean–median difference **prove** that ₹5,500 is an outlier?
 7. Which statistical method would you use to verify whether ₹5,500 is an outlier?
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Q2 — SD vs CV: Employee Productivity

Team	Mean Output	SD
A	600	48
B	2,400	144

Answer:

1. Calculate CV for A.
 2. Calculate CV for B.
 3. Which team has greater absolute variability?
 4. Which team has greater relative variability?
 5. Which team is more consistent relative to its mean?
 6. Why is SD alone not enough for this comparison?
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Q3 — IQR vs Range + Extreme Value

Store A

1000

1100

1200

1300

1400

1500

1600

Store B

1000

1100

1200

1300

1400

1500

6000

Answer:

1. Calculate the range for A and B.
 2. Calculate Q1 for A and B.
 3. Calculate Q3 for A and B.
 4. Calculate IQR for A and B.
 5. Which measure is better for comparing the middle 50%?
 6. Explain why ₹6,000 affects the range much more than the IQR.
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Q4 — Five Number Summary + Skewness

Statistic	Value
Minimum	12
Q1	20
Median	27
Q3	34
Maximum	82

Answer:

1. Calculate the lower whisker length.
2. Calculate the upper whisker length.
3. Which side has the longer tail?
4. Identify the likely skewness.
5. Give two numerical reasons for your conclusion.

6. Can you say that ₹82 is definitely an outlier? Why or why not?

Q5 — Box Plot Business Comparison

Statistic	Store A	Store B
Minimum	₹18,000	₹12,000
Q1	₹28,000	₹22,000
Median	₹35,000	₹38,000
Q3	₹44,000	₹62,000
Maximum	₹52,000	₹1,05,000

Answer:

1. Which store has the higher median?
 2. Calculate IQR for both.
 3. Calculate range for both.
 4. Which store has greater variability?
 5. Which store is more consistent?
 6. Calculate the upper whisker length for both.
 7. Which store has stronger evidence of a long right tail?
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Q6 — Outlier Detection + Business Decision

Delivery Time

10

12

14

15

17

18

20

22

24

50

Using the IQR method:

1. Calculate Q1.
 2. Calculate Q3.
 3. Calculate IQR.
 4. Calculate the lower fence.
 5. Calculate the upper fence.
 6. Identify the outlier.
 7. Should the company use mean or median to represent typical delivery time?
 8. Explain your decision using **outlier + resistance**, not just “median is better.”
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Q7 — CV Ranking + Investment Decision

Investment	Mean Return	SD
A	9%	1.8%
B	16%	3.2%
C	24%	3.6%

Answer:

1. Calculate CV for A.
2. Calculate CV for B.
3. Calculate CV for C.
4. Rank the investments from lowest CV to highest CV.
5. Which investment has the highest return?

6. Which investment has the lowest relative variability?
 7. Does the highest-return investment also have the lowest CV?
 8. If the investor wants **high return + low relative variability**, which investment should be considered first?
 9. Why is CV more useful than SD for this comparison?
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Q8 — Mean vs Median: Don't Overclaim

Platform	Mean Order Value	Median Order Value
A	₹1,650	₹1,050
B	₹1,300	₹1,150

Answer:

1. Which platform has the higher mean?
 2. Which platform has the higher median?
 3. Which platform shows stronger evidence of right-skewness?
 4. Calculate the mean–median gap for both.
 5. Does Platform A's larger gap prove that it contains an outlier?
 6. What method would you use to investigate possible outliers?
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Q9 — Choose the Best Statistical Tool

Choose from:

Mean, Median, SD, IQR, CV, Box Plot, Range

A.

A company wants to report typical employee salary, but a few executives earn extremely high salaries.

B.

Two factories have very different average production levels and management wants to compare relative consistency.

C.

An analyst wants to measure the spread of the central 50% of delivery times.

D.

An analyst wants to compare the distributions of multiple departments and identify possible outliers visually.

E.

Two machines have the same average output and management wants to compare absolute variability.

F.

Management wants the simplest measure from minimum to maximum.

G.

A roughly symmetric dataset has no major extreme values and management wants the arithmetic average.

Q10 — Full Data Analyst Case

Platform	Mean Order Value	Median	SD
A	₹1,600	₹1,100	₹320
B	₹2,100	₹1,950	₹315

Answer:

1. Calculate CV for Platform A.
2. Calculate CV for Platform B.
3. Which platform has greater absolute variability?
4. Which platform has greater relative variability?
5. Which platform shows stronger evidence of right-skewness?
6. Which platform has the higher typical order value?
7. Which platform is more consistent relative to its mean?
8. If management wants higher typical order value, which platform is better?
9. If management wants lower relative variability, which platform is better?
10. Does Platform A's mean–median gap prove that it has an outlier?
11. Give a final analyst conclusion using **mean + median + SD + CV together**.